



**Department of Electrical, Computer
& Biomedical Engineering**
Faculty of Engineering & Architectural Science

Programs: Electrical Engineering

Course Title:	ELE632
Course Number:	Signal and Systems II
Semester/Year (e.g.F2016)	Winter/2026

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Assignment/Lab Number:	3
Assignment/Lab Title:	Discrete-Time Fourier Series

Submission Date	Mar 8, 2026
Due Date	Mar 8, 2026

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Table of Contents

Introduction.....	3
Preperation.....	3
PART A: Discrete-Time Fourier Series.....	4
Question 1.....	4
Question 2.....	4
Question 3.....	6
PART B: Inverse DTFS and time shifting property.....	8
Question 1.....	8
Question 2.....	9
PART C: System Response.....	10
Question 1.....	10
Question 2.....	11
Question 3.....	12
Question 4.....	13
Conclusion.....	15
References.....	15
Appendix.....	16
Part A.....	16
Part B.....	17
Part C.....	18

INTRODUCTION

In this lab, students will have an opportunity to learn about the Discrete-Time Fourier Series (DTFS) through the creation of a program in Python that implements the polyphase decomposition of the DTFS, as well as learning about the properties associated with the DTFS.

Previously, in Lab 1, recursion methods were used to determine the physical attributes (such as transformations, energy, and power) of discrete signals using algorithms and data structures in the Python programming language. In lab 2, the basic concepts of discrete-time signal processing to analyze mathematical models and digital implementations to calculate the unit impulse response, zero-input response, zero-state response, and total response for linear time-invariant discrete-time (LTID) systems.

Throughout these experiments, the following student number will be computed as per the given sequence of variables.

$$\begin{aligned} \text{Student ID} &= 501103255 & [0] \\ [A = 5, B = 0, C = 1, D = 1, E = 0, F = 3, G = 2, H = 5, I = 5] \end{aligned}$$

Equation 0: Assigned Student Number Sequence

PREPERATION

The Discrete-time Fourier series D_r for a periodic signal $x[n]$ with the fundamental period N_0 is defined as follows:

$$x[n] = \sum_{r=0}^{N_0-1} D_r e^{jr\Omega_0 n} \quad [1]$$

Equation 1: Fourier Series

$$D_r = \frac{1}{N_0} \sum_{n=0}^{N_0-1} x[n] e^{-jr\Omega_0 n} \quad [2]$$

Equation 2: Fourier Series Coefficients

$$\Omega_0 = \frac{2\pi}{N_0} \quad [3]$$

Equation 3: Fundamental Frequency

PART A: Discrete-Time Fourier Series (DTFS)

Question 1: $x[n] = 4\sin(2.4\pi n) + 2\cos(3.2\pi n)$

In the given equation $\omega_1 = 2.4\pi$ and $\omega_2 = 3.2\pi$. We also know that $\frac{\omega_0}{2\pi} = \frac{m}{N}$ and $N = N_0$ if $\frac{m}{N}$ is in its lowest form and is rational. Furthermore, it can be concluded that the signal is periodic if the ratio of $\frac{m}{N}$ which equals $\frac{\omega_0}{2\pi}$ is rational for each term given in the signal. It should be noted that rationality refers to an expression represented as a ratio of whole numbers.

$$\frac{\omega_1}{2\pi} = \frac{2.4\pi}{2\pi} = \frac{2.4}{2} = \frac{6}{5} \text{ which is rational and } N_{0_1} = 5.$$

$$\frac{\omega_2}{2\pi} = \frac{3.2\pi}{2\pi} = \frac{3.2}{2} = \frac{16}{10} = \frac{8}{5} \text{ which is rational and } N_{0_2} = 5.$$

Thus, the signal $x[n]$ is periodic since both terms (constituent sums of signals) are periodic and we can compute the fundamental period, N_0 :

$$\begin{aligned} N_0(x[n]) &= LCM(N_{0_1}, N_{0_2}) = LCM(5, 5) = 5 \\ N_0 &= 5 \\ \Omega_0 &= \frac{2\pi}{N_0} = \frac{2\pi}{5} = \frac{2\pi}{5} \end{aligned}$$

Therefore, the fundamental period, $N_0 = 5$ and the fundamental frequency, $\Omega_0 = \frac{2\pi}{5}$ for $x[n]$.

Question 2:

Equations [1] & [3] above (in the preparation section) can be compared with Equation 9.4 from the textbook as illustrated below:

Let us consider the first choice, which corresponds to exponentials $e^{jr\Omega_0 n}$ for $r = 0, 1, 2, \dots, N_0 - 1$. The Fourier series for an N_0 -periodic signal $x[n]$ consists of only these N_0 harmonics, and can be expressed as

$$x[n] = \sum_{r=0}^{N_0-1} D_r e^{jr\Omega_0 n} \quad \Omega_0 = \frac{2\pi}{N_0} \quad (9.4)$$

Figure 0: Linear Systems and Signals (B. P. Lathi)^[1]

Using the NumPy's FFT (Fast Fourier Transform) function, the Discrete-Time Fourier Series (DTFS) was computed. Then, following the computation given below, the Magnitude and Phase Spectra plots were generated. Note the range of n had to be changed from -10 to 10 to 1 to 20 to accommodate the magnitude and phase spectra plot generation.

```
x_vals = x(n)
X = np.fft.fft(x_vals)
```

Image 0: Fast Fourier Transform (FFT) algorithm

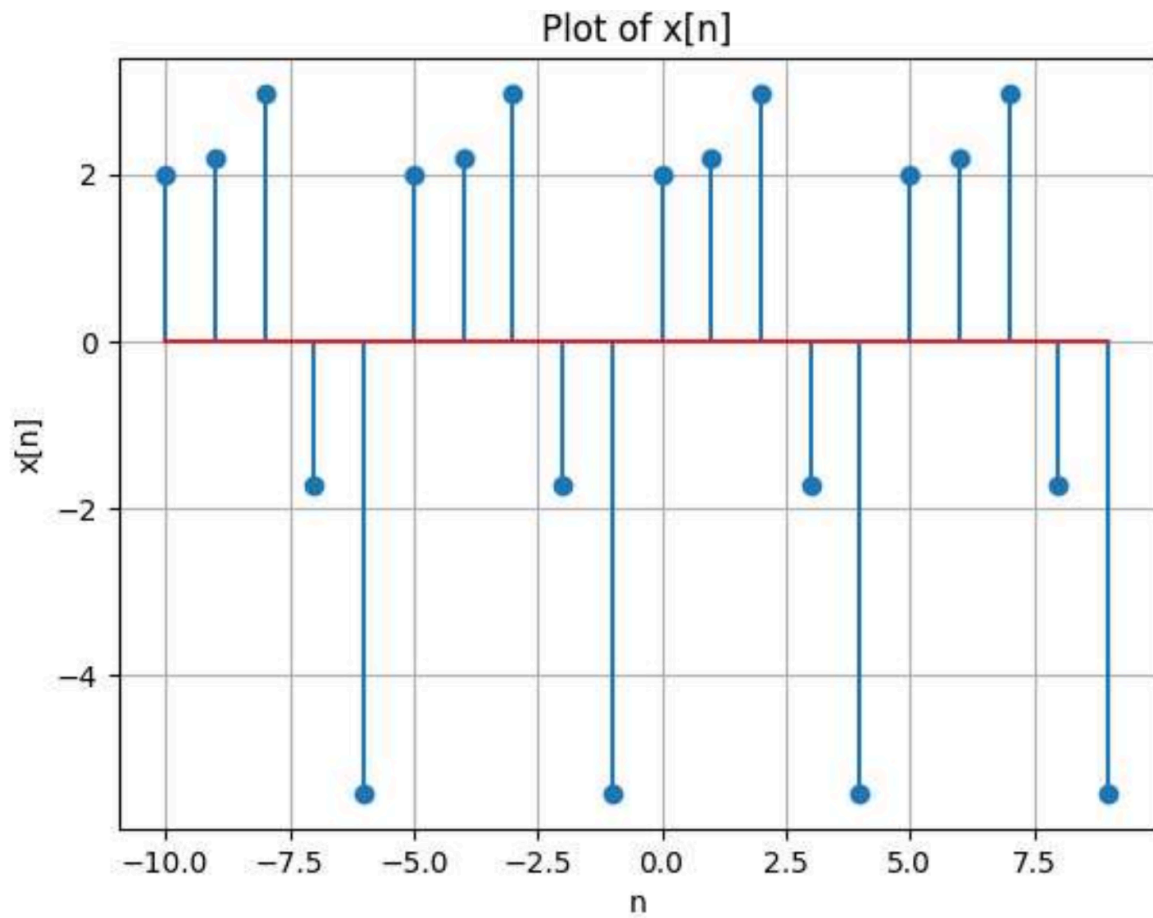


Figure 1: Plot of the Discrete-Time Fourier Series (DTFS) for $x[n]$

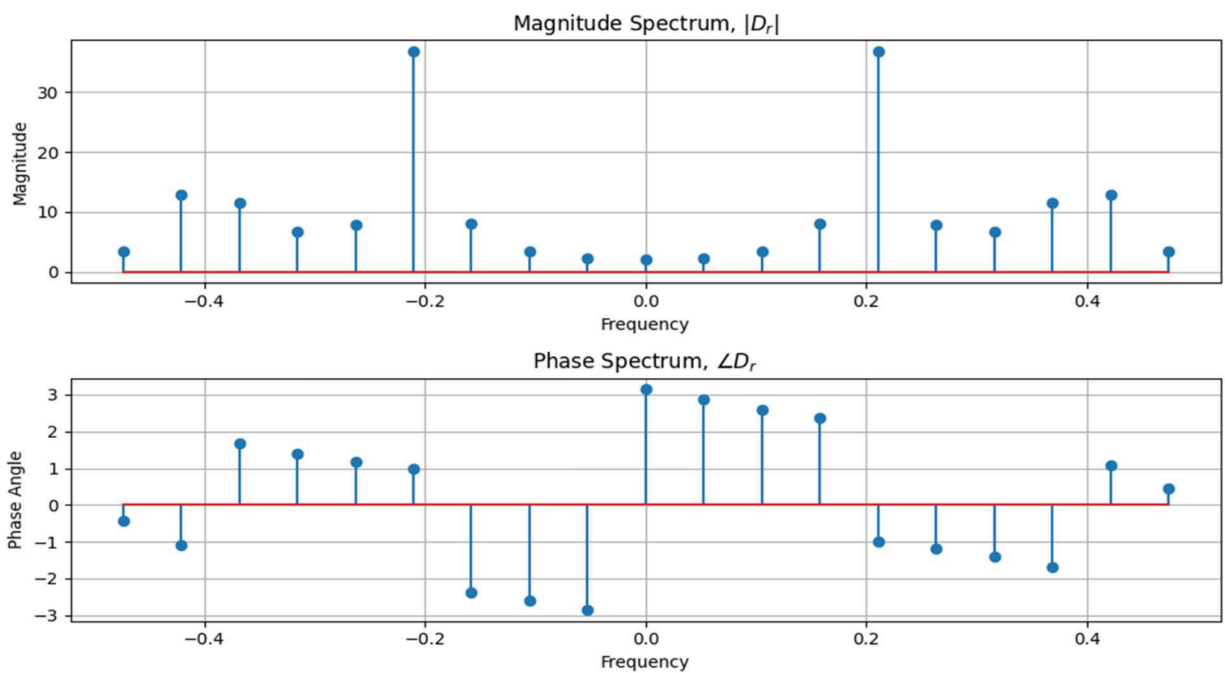


Figure 2: Magnitude and Phase Spectrum for $x[n]$

Question 3:

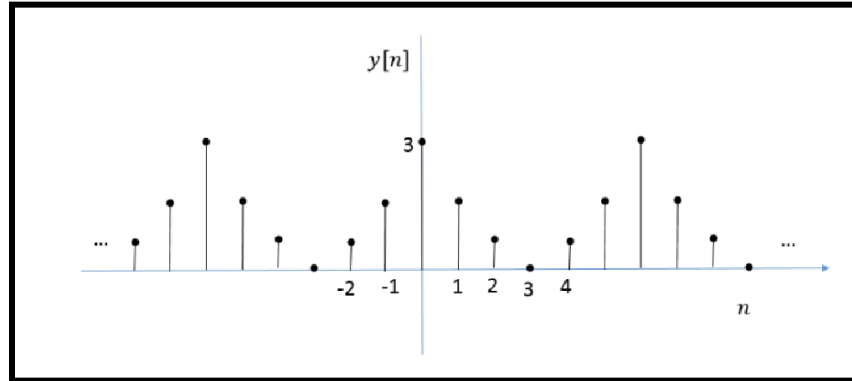


Image 1: Signal of $y[n]$

- I. The fundamental period (N_0) and the fundamental frequency (Ω_0) of $y[n]$

By simply looking at the signal, $y[n]$ it is clear that the signal is periodic since it repeats its exact shape at consistent intervals (periods) over time, meaning the graph looks identical if shifted by a multiple of that interval, for which, in this case starting at zero. Thus, the fundamental period, N_0 of the signal is 6 units, $N_0 = 6$.

We also know that $T = \frac{2\pi}{\omega} = 6$, and by rearranging for ω (the fundamental angular frequency) $\omega = \frac{2\pi}{T} = \frac{2\pi}{6} = \frac{\pi}{3}$. The value of $\omega = \omega_0$ can be compared with $\frac{\omega_0}{2\pi} = \frac{m}{N}$, which determines if the signal is periodic based on rationality.

$\frac{\frac{\pi}{3}}{2\pi} = \frac{1}{6}$ which is rational and thus $N = N_0 = 6$ since the signal of $y[n]$ is periodic.

$$N_0 = T = 6$$

$$\Omega_0 = \frac{2\pi}{N_0} = \frac{2\pi}{6} = \frac{\pi}{3}$$

Therefore, the fundamental period, $N_0 = 6$ and the fundamental frequency, $\Omega_0 = \frac{\pi}{3}$ for $y[n]$.

II. Discrete-Time Fourier Series (DTFS) of signal $y[n]$, the magnitude $|D_r|$, and phase $\angle D_r$

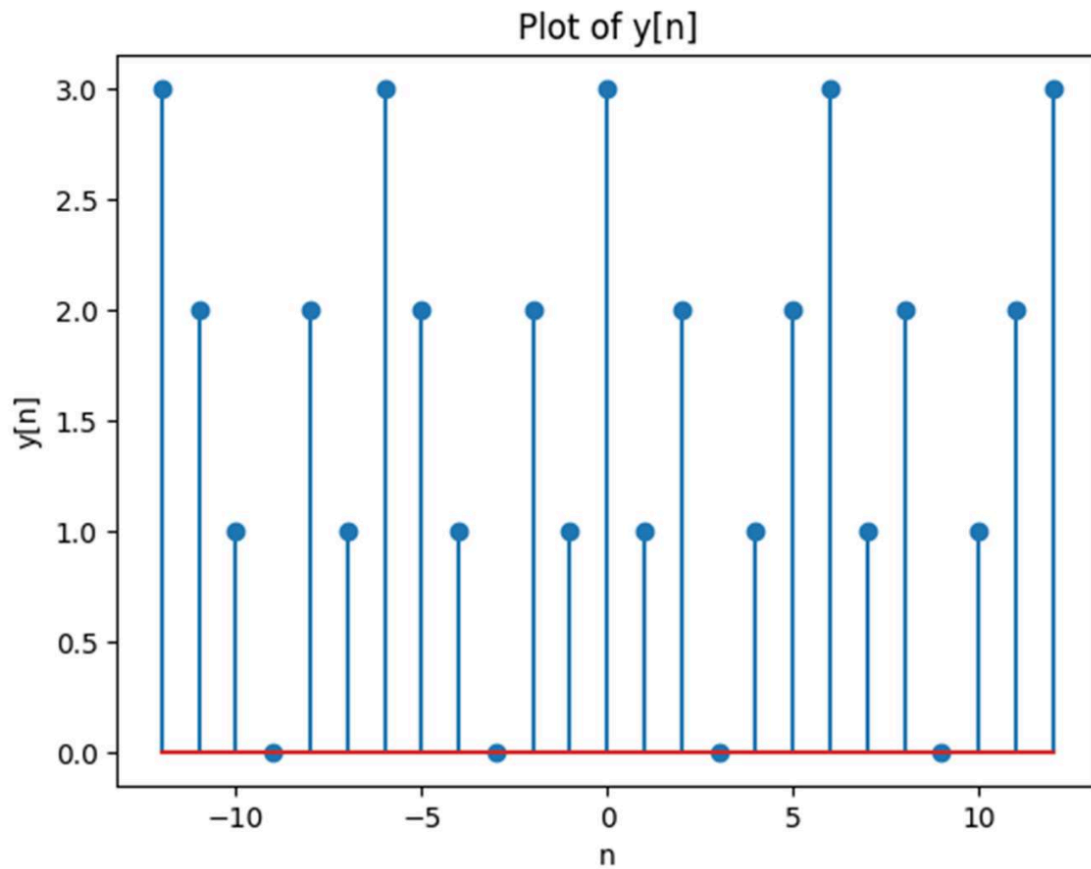


Figure 3: Plot of the Discrete-Time Fourier Series for $y[n]$

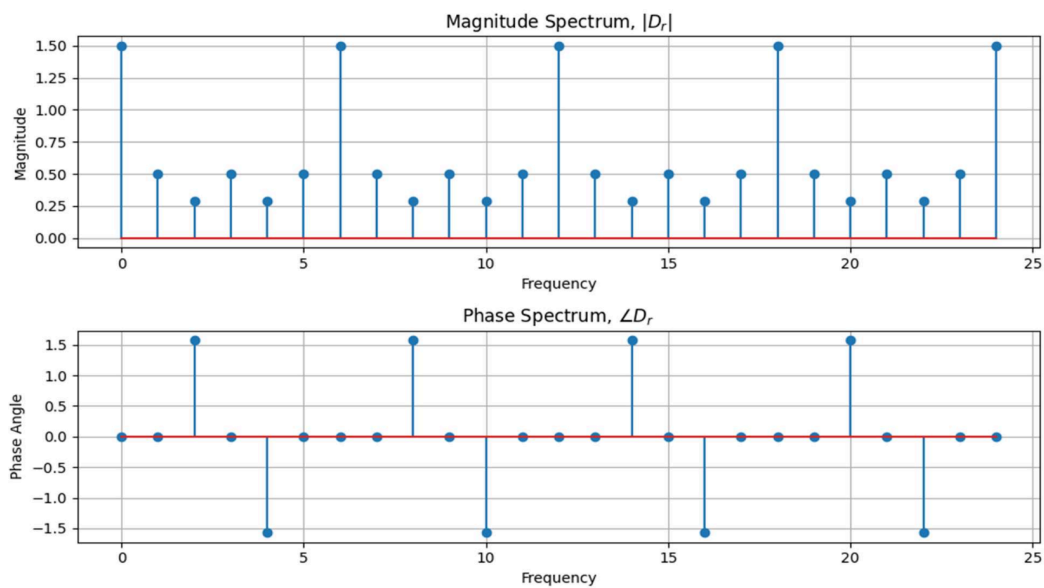


Figure 4: Magnitude ($|D_r|$) and Phase ($\angle D_r$) Spectrum for $y[n]$

PART B: Inverse Discrete-Time Fourier Series (DTFS) and time shifting property

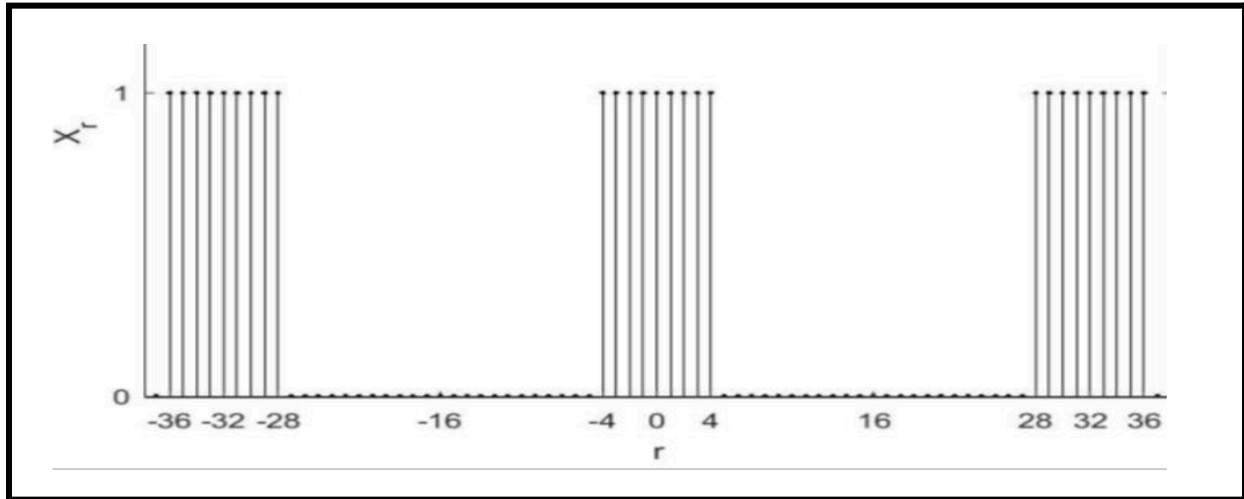


Image 2: Fourier Series Spectrum, X_r for a periodic discrete signal, $x[n]$

Question 1:

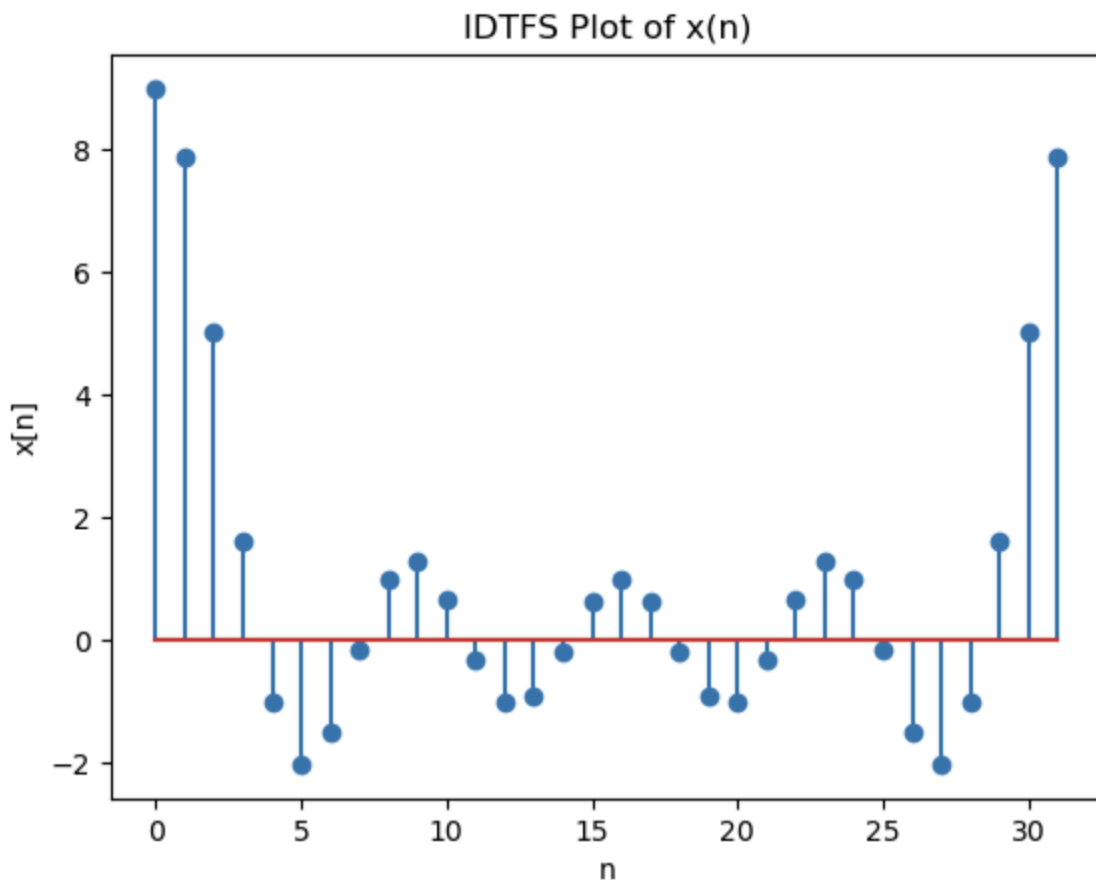


Figure 5: Plot of the X_r Inverse Discrete-Time Fourier Series (IDTFS) for $x[n]$, $F^{-1}(X_r) = x[n]$

Question 2:

Multiplying $X[r]$ to $e^{-j(D+1)\Omega_0 r}$ where $D = 1$. Note that the fundamental period can be derived from the inverse plot of the Fourier spectra ($X[r]$), which is the discrete-time Fourier series of the signal $x[n]$. Thus, counting the units of n for consistent intervals referenced from Figure 5, N_0 was found to be 32 as the fundamental period, and thus $\Omega_0 = \frac{\pi}{16}$.

$$X[r] \times e^{-j(D+1)\Omega_0 r} = X[r] \times e^{-j2\frac{\pi}{16}r} = X[r] \times e^{-j\frac{\pi}{8}r}$$

As compared to the plot of $x[n]$ referenced in part 1 (Figure 5), representing the inverse discrete time fourier series of $X[r]$, this result of multiplying $X[r]$ to $e^{-j\frac{\pi}{8}r}$ has the effect of shifting the signal 2 units to the right.

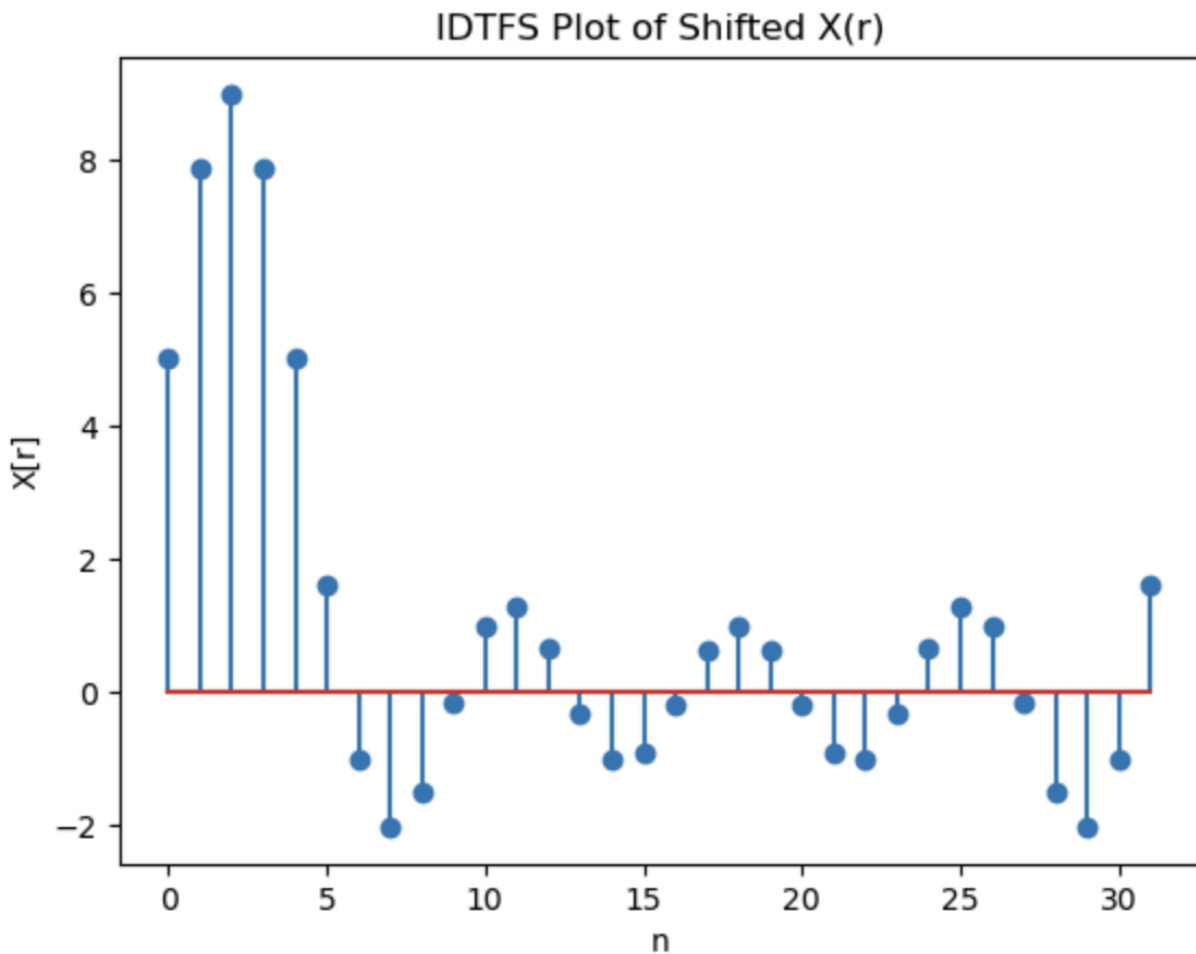


Figure 6: Plot of $x[n]$ following multiplication in the frequency domain

PART C: System Response

The system response will compute the response of a Linear Time-Invariant Discrete-time (LTID) system to an input signal using Discrete-Time Fourier Series (DTFS).

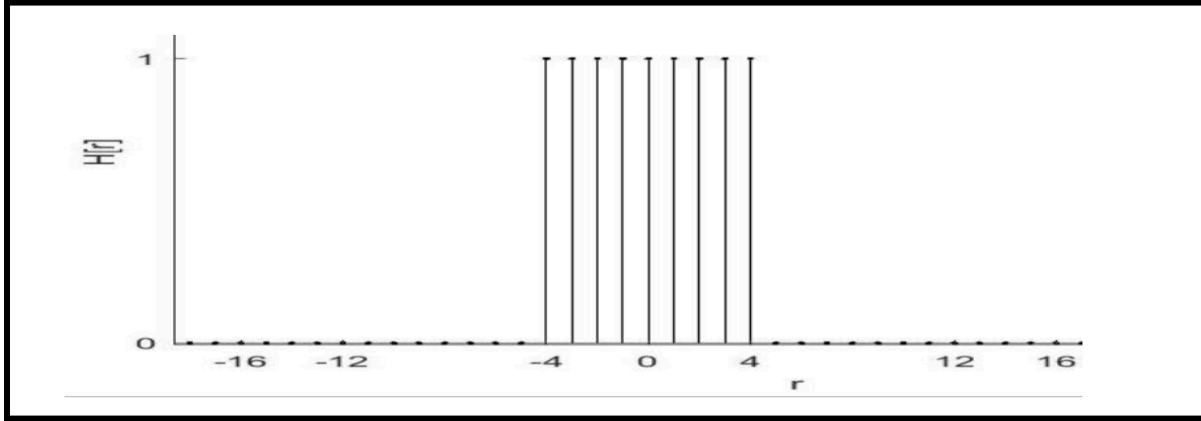


Image 3: Fourier Series Spectrum of $x[n]$ (a periodic discrete signal)

Question 1:

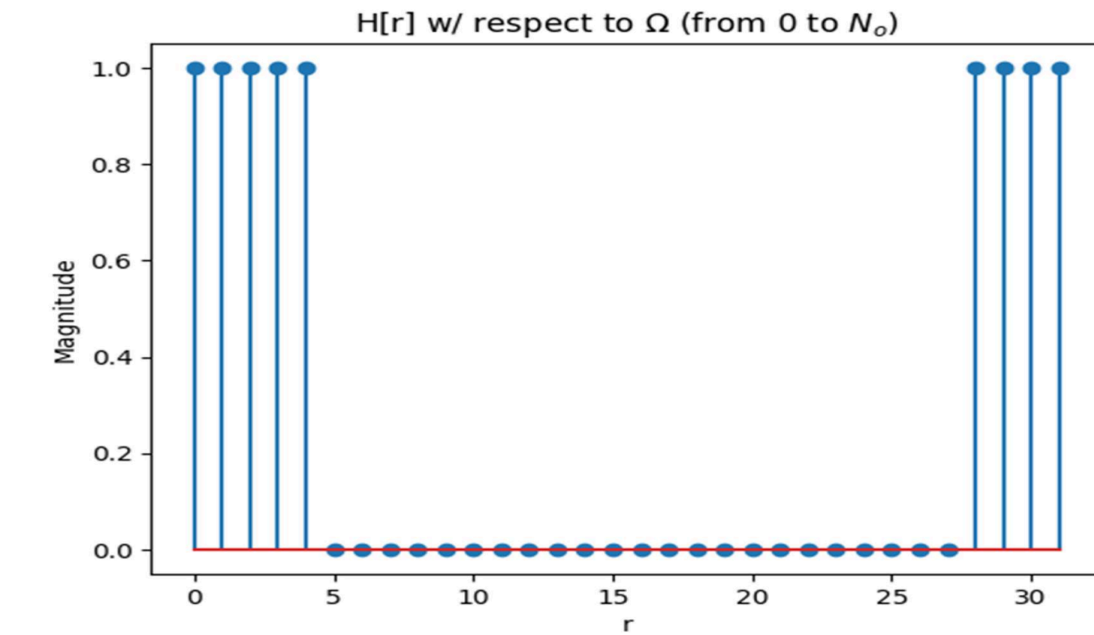


Figure 7: Plot of $H[r]$ with respect to Ω_0

The frequency response of an LTID (Linear Time-Invariant Discrete-time) system, H_r , was referenced from the lab manual, Image 3. Note that $H[r]$ was plotted with respect to Ω_0 . H_r has a fundamental frequency $N_0 = 32$ and as a result, $\Omega_0 = \frac{2\pi}{32} = \frac{\pi}{16}$.

Question 2: $x_1[n] = 4\cos(\frac{\pi n}{8})$

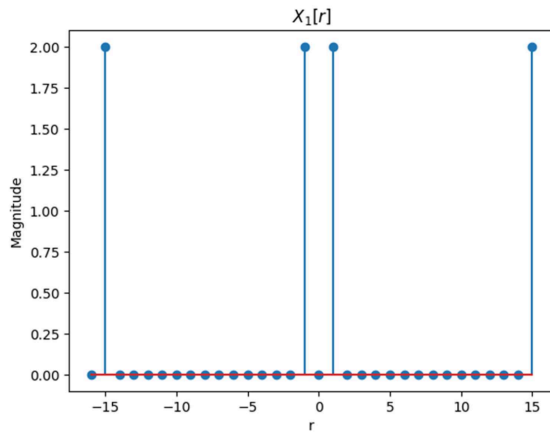


Figure 8: DTFS of $x_1[n]$

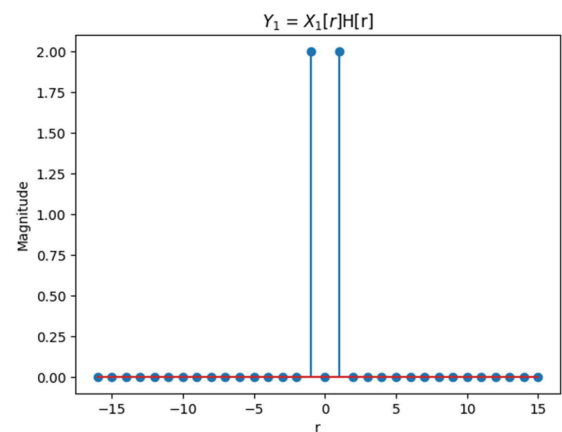


Figure 9: $Y_1[r] = X_1[r]H[r]$

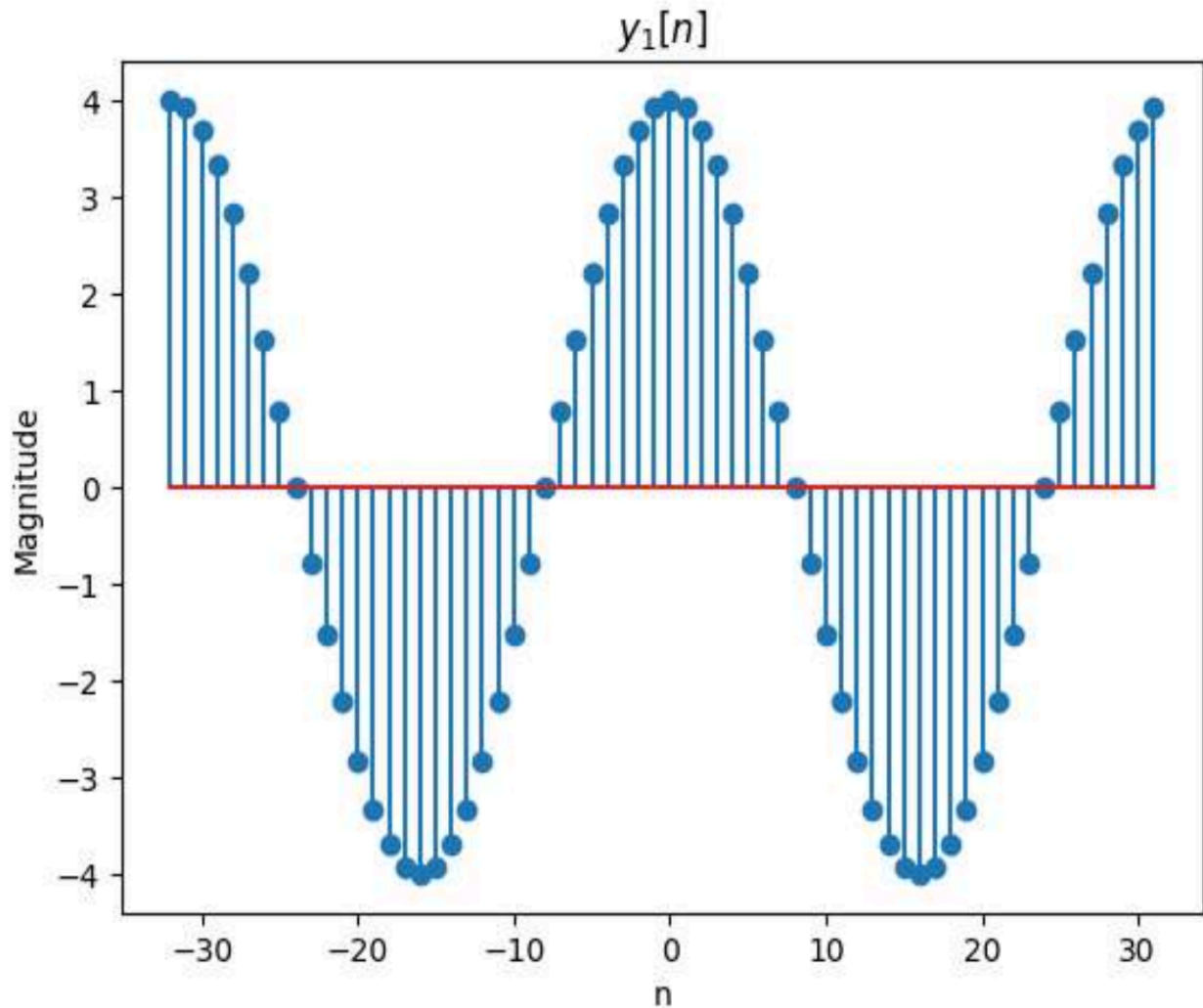


Figure 10: Inverse Discrete-Time Fourier Series (DTFS) to find the response to the input, $y_1[n]$

Question 3: $x_2[n] = 4\cos(\frac{\pi n}{3})$

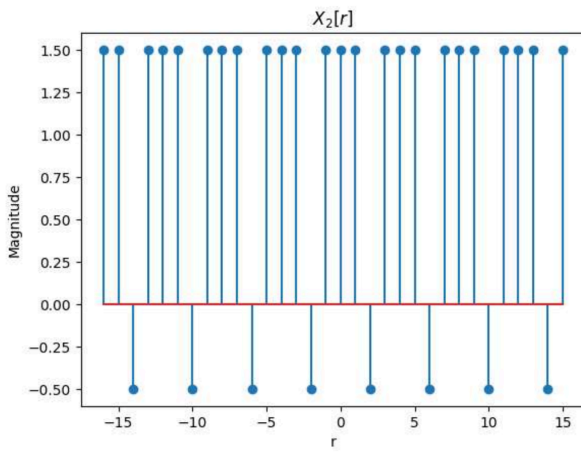


Figure 11: DTFS of $x_2[n]$

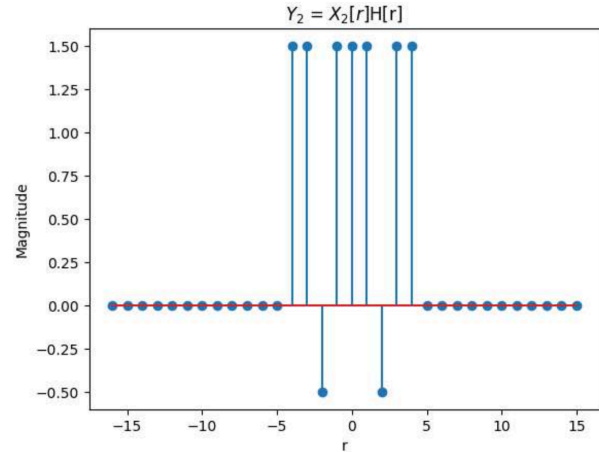


Figure 12: $Y_2[r] = X_2[r]H[r]$

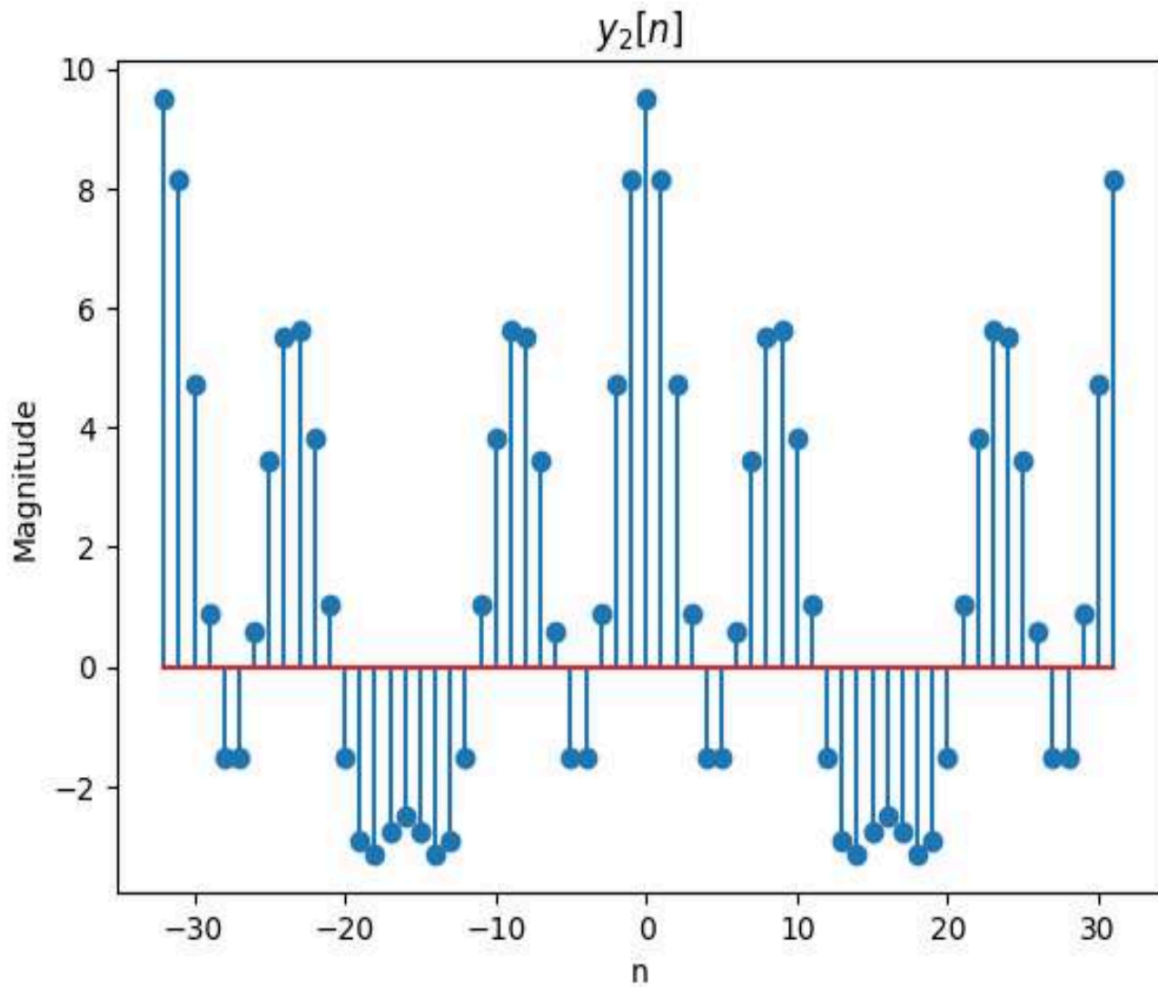


Figure 13: Inverse Discrete-Time Fourier Series (DTFS) to find the response to the input, $y_2[n]$

Question 4:

The graphs differ because the frequencies differ. The filter H_R is a Discrete-Time Low-Pass Filter with a passband of $[-4, 4]$. As they are different, the cosine portions within Figures 10 & 13 are of varying frequencies.

<u>Feature</u>	<u>Question 2</u>	<u>Question 3</u>
	$y_1[n]$	$y_2[n]$
Input Signal	$x_1[n] = 4\cos(\frac{\pi n}{8})$	$x_2[n] = 4\cos(\frac{\pi n}{3})$
Input Frequency	<i>Low</i> ($\pi/8$)	<i>High</i> ($\pi/3$)
Spectral Location	Harmonics at $r = \pm 1$	Harmonics spread widely across the spectrum
Filter Interaction	• <i>Fully within the passband</i> The harmonics at ± 1 are inside the $[-4, 4]$ range.	• <i>Partially blocked</i> Harmonics outside the $[-4, 4]$ range are deleted.
Resulting Shape	• <i>Pure Sinusoid</i> The signal is preserved and remains a clean cosine wave.	• <i>Complex Periodic Waveform</i> The signal is distorted (decayed) and modified, losing its original cosine shape.

$H[R]$ is a Discrete-Time Low-Pass Filter with a passband from $r = -4$ to $r = 4$, as seen in Image 3. A frequency component is multiplied by 1 (passes through) if it lies inside $[-4, 4]$. A frequency component is multiplied by 0 (is removed) if it is outside of $[-4, 4]$.

For $x_1[n] = 4\cos(\frac{\pi n}{8})$, the fundamental frequency is $\omega_1 = \frac{\pi}{8}$. As seen in Figure 8, the spikes (harmonics) in the DTFS domain ($X_1[r]$) are situated at $r = \pm 1$. The signal passes through entirely unaltered because $r = \pm 1$ comfortably lies inside the filter's passband ($[-4, 4]$). As a result, $Y_1[r]$ resembles $X_1[r]$ identically. Therefore, $y_1[n]$ continues to be a perfect cosine wave with an amplitude of 4.

For $x_2[n] = 4\cos(\frac{\pi n}{3})$, the fundamental frequency is $\omega_1 = \frac{\pi}{3}$. The components of this signal's DTFS ($X_2[r]$) are substantially farther apart. Figure 11 shows a number of spikes throughout the spectrum. Only the spikes at $r = 0, \pm 1, \pm 2, \pm 3, \pm 4$ are permitted to pass through the filter $H[r]$. Any elements of $x_2[n]$ at $|r| > 4$ are eliminated. As a result, the original spectrum is truncated in $Y_2[r]$ (Figure 12). The result $y_2[n]$ is a warped, smoothed version of the input, having eliminated high-frequency components (the clarity of the signal). The filter significantly altered the signal's harmonic content, making it appear like a complicated, periodic oscillation rather than a clear cosine wave.

Frequency Selectivity is one of the key concepts involved in different output signals (cosine signals) for two different reasons, the first reason being the different Signal Properties of the two Filters; specifically, Filter $H[R]$ is considered the Gatekeeper of the signal's frequency components (it decides if a particular frequency component will be allowed to pass through or not).

The filter leaves $y_1[n]$ untouched because its frequency is low enough to pass through entirely. Conversely, $y_2[n]$ is significantly distorted because the filter cuts off its high-frequency components, fundamentally altering its harmonic structure.

CONCLUSION

The objective of this experiment was to use Python to effectively create and study the Discrete-Time Fourier Series (DTFS) and its related features. Several important signal processing concepts were confirmed by the creation of methods that calculate both the forward and inverse DTFS.

Periodicity and spectral representation were shown in Part A that discrete frequency components might be extracted from complicated periodic signals. The display of magnitude and phase spectra made possible by the use of NumPy's FFT function confirmed that a signal's information is frequently better comprehended in the frequency domain.

Time-Shifting Property of the DTFS's shifting property was verified in Part B. The time-domain signal $x[n]$ was effectively shifted by multiplying the frequency-domain coefficients by a complex exponential $e^{-j(D+1)\Omega_0 r}$, demonstrating the clear connection between phase shifts in frequency and delays in time.

System Response and Frequency Selectivity implies that the idea of frequency selectivity was emphasized in the last tests in Part C. It was found that the output of a Discrete-Time Low-Pass Filter $H[r]$ is only dependent on the spectral location of the signal with respect to the filter's passband. The low-frequency signal $x_1[n]$ went through the filter (passband $[-4, 4]$) nearly unchanged. The major harmonics of the higher-frequency signal $x_2[n]$ were blocked, causing considerable distortion and a loss of the original sinusoidal integrity.

In conclusion, this lab gave me a thorough understanding of how periodic signals interact with LTI systems. An essential capability for electrical engineers is the capacity to modify signals in the frequency domain. This allows them to create systems and filters that can isolate or remove particular signal components according to their frequency characteristics.

REFERENCES

[1] Lathi, B. P., & Green, R. A. (2018). Linear Systems and Signals (3rd ed.). Oxford University Press.

APPENDIX

PART A

Question 1:

```
import numpy as np
import matplotlib.pyplot as plt
n = np.arange(0, 21)
x = 4*np.sin(2.4*np.pi*n) + 2*np.cos(3.2*np.pi*n)
mN1 = (2.4*np.pi)/(2*np.pi)
mN2 = (3.2*np.pi)/(2*np.pi)
#m/N1 = 6/5
#m/N2 = 8/5
N0 = 5
Om0 = 2*np.pi/N0
```

Question 2:

```
import numpy as np
import matplotlib.pyplot as plt
n = np.arange(0, N0)
x = 4*np.sin(2.4*np.pi*n) + 2*np.cos(3.2*np.pi*n)
Dr = np.zeros(N0, dtype=complex)
for r in range(N0):
    Dr[r] = np.sum(x * np.exp(-1j * r * n * Om0)) / N0

plt.stem(n, x)
plt.title('DTFS Plot of x(n)')
plt.xlabel('n')
plt.ylabel('x[n]')
plt.show()

plt.stem(np.abs(Dr))
plt.title('|x(n)| DTFS')
plt.xlabel('r')
plt.ylabel('|Dr|')
plt.show()

plt.stem(np.angle(Dr))
plt.title('<x(n) DTFS')
plt.xlabel('r')
plt.ylabel('<Dr')
plt.show()
```

Question 3-ii:

```
import numpy as np
import matplotlib.pyplot as plt

N0 = 6
Om0 = 2*np.pi/N0

n = np.arange(0, N0)
```

```

y = np.array([3, 2, 1, 0, 1, 2])
Dr = np.zeros(N0, dtype=complex)
for r in range(N0):
    Dr[r] = np.sum(y * np.exp(-1j * r * n * Om0)) / N0

plt.stem(n, y)
plt.title('DTFS Plot of y(n)')
plt.xlabel('n')
plt.ylabel('y[n]')
plt.show()

plt.stem(np.abs(Dr))
plt.title('|y(n)| DTFS')
plt.xlabel('r')
plt.ylabel('|Dr|')
plt.grid(True)
plt.show()

plt.stem(np.angle(Dr))
plt.title('<y(n) DTFS')
plt.xlabel('r')
plt.ylabel('<Dr')
plt.grid(True)
plt.show()

```

PART B

Question 1:

```

import numpy as np
import matplotlib.pyplot as plt

N0 = 32
Om0 = 2*np.pi/N0
n = np.arange(N0)

xr = np.concatenate((np.ones(5), np.zeros(23), np.ones(4)))
x = np.fft.ifft(xr)*N0

plt.stem(n, x)
plt.title('IDTFS Plot of x(n)')
plt.xlabel('n')
plt.ylabel('x[n]')
plt.show()

```

Question 2:

#NOTE: My D value is 1

```
import numpy as np
import matplotlib.pyplot as plt
```

```
N0 = 32
Om0 = 2*np.pi/N0
n = np.arange(N0)

xr = np.concatenate((np.ones(5), np.zeros(23), np.ones(4)))
X = xr*np.exp(-2j*1*n*Om0)
X = np.fft.ifft(X)*N0
```

```
plt.stem(n, X)
plt.title('IDTFS Plot of Shifted X(r)')
plt.xlabel('n')
plt.ylabel('X[r]')
plt.show()
```

The exponential term shifts the spectrum left or right depending on the D value (or n in general).

PART C

Question 1:

```
import numpy as np
import matplotlib.pyplot as plt
```

```
N0 = 32
Om0 = 2*np.pi/N0
r = np.arange(N0)
Hr = np.concatenate((np.ones(5), np.zeros(23), np.ones(4)))
```

```
plt.stem(Om0*r, Hr)
plt.title('H[r] wrt Omega0')
plt.xlabel('r')
plt.ylabel('H[r]')
plt.show()
```

Question 2:

```
import numpy as np
import matplotlib.pyplot as plt

N0 = 32
Om0 = 2 * np.pi / N0
r = np.arange(N0)

Hr = np.concatenate((np.ones(5), np.zeros(23), np.ones(4)))
x = 4 * np.cos(np.pi * r / 8)
X = np.fft.fft(x)

fig, axs = plt.subplots(2, 1, figsize=(8, 8))

axs[0].stem(r, X.real)
axs[0].set_title('Plot of X[r]')
axs[0].set_xlabel('n')
axs[0].set_ylabel('X[r]')

Y = X * Hr
y = np.fft.ifft(Y)

axs[1].stem(r, y.real)
axs[1].set_title('Plot of y[n]')
axs[1].set_xlabel('n')
axs[1].set_ylabel('y[n]')

plt.tight_layout()
plt.show()
```

Question 3:

```
import numpy as np
import matplotlib.pyplot as plt

N0 = 32
Om0 = 2*np.pi/N0
r = np.arange(0, N0)

Hr = np.concatenate((np.ones(5), np.zeros(23), np.ones(4)))
x = 4*np.cos(np.pi*r/2)
X = np.fft.fft(x)

fig, ax = plt.subplots()
ax.stem(r, X)
ax.set_title('Plot of X[r]')
ax.set_xlabel('n')
ax.set_ylabel('X[r]')

Y = X*Hr
y = np.fft.ifft(Y)

fig, ax = plt.subplots()
ax.stem(r, y.real)
ax.set_title('Plot of y\u2082082[n]')
ax.set_xlabel('n')
ax.set_ylabel('y[n]')
```